22.2 Wave-Particle Duality

Question Paper

Course	CIE A Level Physics
Section	22. Quantum Physics
Topic	22.2 Wave-Particle Duality
Difficulty	Medium

Time allowed: 30

Score: /23

Percentage: /100

Question 1a

State the formula for the de Broglie wavelength λ of a moving particle.

State the meaning of any other symbol used.

[2 marks]

Question 1b

Electrons accelerate through a potential difference, pass through a thin crystal and are then incident on a fluorescent screen.

The pattern in Fig. 1.1 is observed on the fluorescent screen.

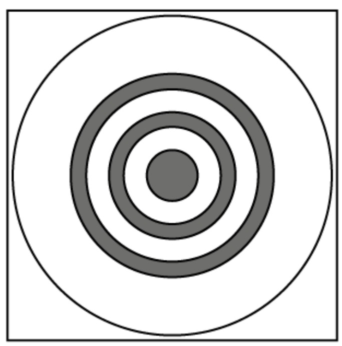


Fig. 1.1 not to scale

(i)
State the name of the phenomenon shown by the electrons at the crystal.

[1]

(ii)
State what this phenomenon shows about the nature of electrons.

[1]

(iii)
Suggest why the thin crystal causes the phenomenon in **(b)(i)**.

[1]

[3 marks]

Question 1c

The potential difference used to accelerate the electron is changed.

The new pattern observed on the screen is shown in Fig. 1.2.

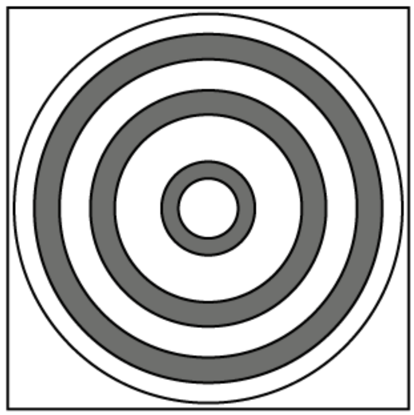


Fig 1.2

State and explain the change that has been made to the potential difference to create the pattern shown in Fig. 1.2.

[2 marks]

Question 2a

State **one** piece of experimental evidence for:

(i)
the wave nature of matter

[1]

(ii)
the particulate nature of electromagnetic radiation.

[1]

[2 marks]

Question 2b

Calculate the de Broglie wavelength λ of an alpha-particle moving at a speed of 5.9×10^7 m s^{-1}.

λ = ... m

[3 marks]

Question 2c

The speed v of the alpha-particles in **(b)(i)** is gradually reduced to zero.

On Fig. 1.1, sketch the variation with v of λ.

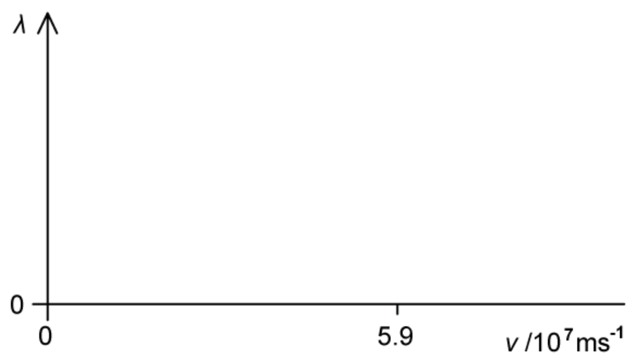


Fig 1.1

[2 marks]

Question 3a

Explain what is meant by the de Broglie wavelength.

[1 mark]

Question 3b

A gamma-ray photon of energy 2.6×10^{-18} J is incident on an isolated stationary electron, as illustrated in Fig. 1.1.

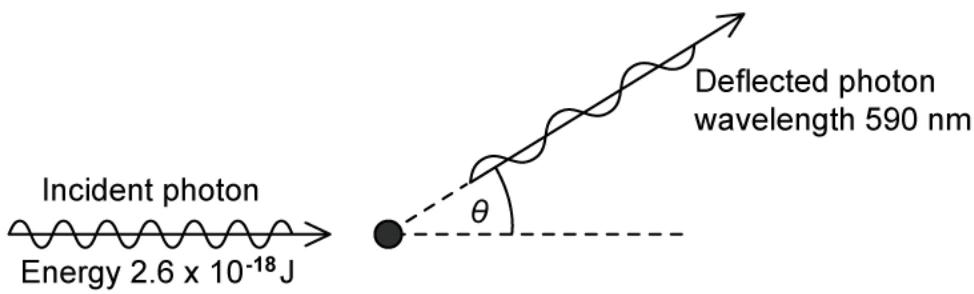


Fig. 1.1

The photon is deflected elastically by the electron through angle θ. The deflected photon has a wavelength of 590 nm.

(i)
On Fig. 1.1, draw an arrow to indicate a possible initial direction of motion of the electron after the photon has been deflected
[1]

(ii)
Calculate the energy of the deflected photon.

photon energy = ... J [2]

[3 marks]

Question 3c

Calculate the speed of the electron after the photon has been deflected.

speed = ... m s^{-1}

[3 marks]

Question 3d

Explain why the magnitude of the final momentum of the electron is not equal to the change in magnitude of the momentum of the photon.

[2 marks]

Model Answers: Medium

1a

(a) The formula for the de Broglie wavelength λ of a moving particle is:

Any **one** pair from:

- $\lambda = \frac{h}{p}$ [1 mark]

Where

- h is the Planck constant **AND** p is the momentum (of the particle); [1 mark]

OR

- $\lambda = \frac{h}{mv}$ [1 mark]

Where:

- m is the mass of the particle **AND** v is the velocity of the particle; [1 mark]

[Total: 2 marks]

1b

(b)

(i) The phenomenon shown by the electrons at the crystal is called:

- Electron diffraction; [1 mark]

(ii) This phenomena shows that:

- Moving electrons behave like waves; [1 mark]

(iii) The thin crystal causes the phenomenon in **(b)(i)** because:

Any **one** from:

- The spacing between the atoms ≈ the wavelength of the electron; [1 mark]
- The diameter of the atom ≈ the wavelength of the electron; [1 mark]

[Total: 3 marks]

1c

(c) The change that has been made to the potential difference to create the pattern shown is:

- The wavelength has decreased **OR** the electron had a lower momentum; [1 mark]
- So the accelerating p.d. was decreased; [1 mark]

[Total: 2 marks]

Decreasing the **p.d. decreases** the **energy** of the electrons incident on the screen

$$\lambda \uparrow = \frac{h}{\sqrt{2mE \downarrow}}$$

Hence, the **wavelength increases**.

The **pattern** observed on the screen is **different**, so the **fringe spacing** is **different**.

$$\lambda = \frac{ax}{D} \text{ and } \frac{\lambda \uparrow D}{a} = x \uparrow$$

From **Young's Double slit equation** the fringe spacing, ***x*** **increases**. Hence, the **rings** on the screen are **further apart**.

2a

(a)

(i) One piece of evidence for the wave nature of matter is:

- Electron diffraction; [1 mark]

(ii) One piece of evidence for the particulate nature of electromagnetic radiation is:

- Photoelectric effect; [1 mark]

[Total: 2 marks]

2b

(b) Calculate the de Broglie wavelength λ of an alpha-particle moving at a speed of 5.9×10^7 m s^{-1}.

List the known quantities:

- An α-particle contains two protons and two neutrons, so has 4 nucleons
- Mass of α-particle, $m = 4u = 4 \times (1.66 \times 10^{-27}) = 6.64 \times 10^{-27}$ kg
- Speed of α-particle, $v = 5.9 \times 10^7$ m s^{-1}
- Planck's constant, $h = 6.63 \times 10^{-34}$ J s

Calculate the momentum of the alpha particle:

- $p = mv$
- $p = (6.64 \times 10^{-27}) \times (5.9 \times 10^7)$
- $p = 3.92 \times 10^{-19}$ N s [1 mark]

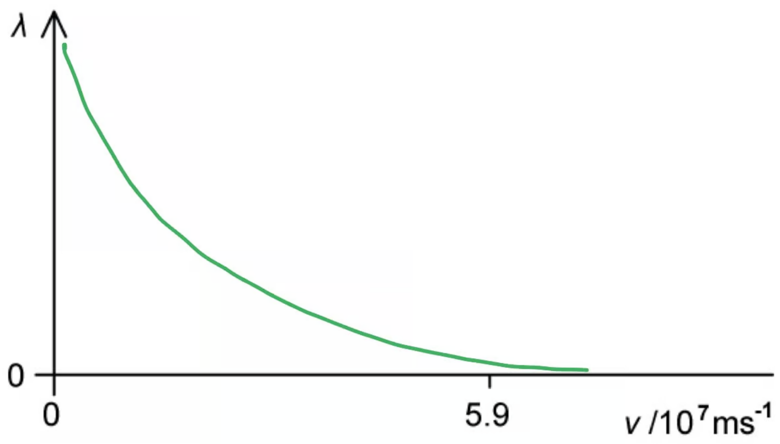
Use the de Broglie equation:

- $\lambda = \frac{h}{p}$ [1 mark]
- $\lambda = \frac{6.63 \times 10^{-34}}{3.92 \times 10^{-19}}$
- $\lambda = 1.69 \times 10^{-15}$ m [1 mark]

[Total: 3 marks]

2c

(c) A sketch of the variation with v of λ that would score 2 marks should look like this:



- The line drawn has a negative gradient throughout; [1 mark]
- The curve does not touch either axes with $\lambda \neq 0$ for $v = 5.9 \times 10^7$ m s^{-1}; [1 mark]

[Total: 2 marks]

Use **equations** you **know** to figure out their **graphical relationship.**

For the de Broglie equation: $\lambda \uparrow = \dfrac{h}{mv \downarrow}$

- When $v \rightarrow 0$
- Then $\lambda \rightarrow \infty$

Hence, the curve cannot touch the λ or v axis.

3a

(a) The de Broglie wavelength is:

- The wavelength associated with a moving particle; [1 mark]

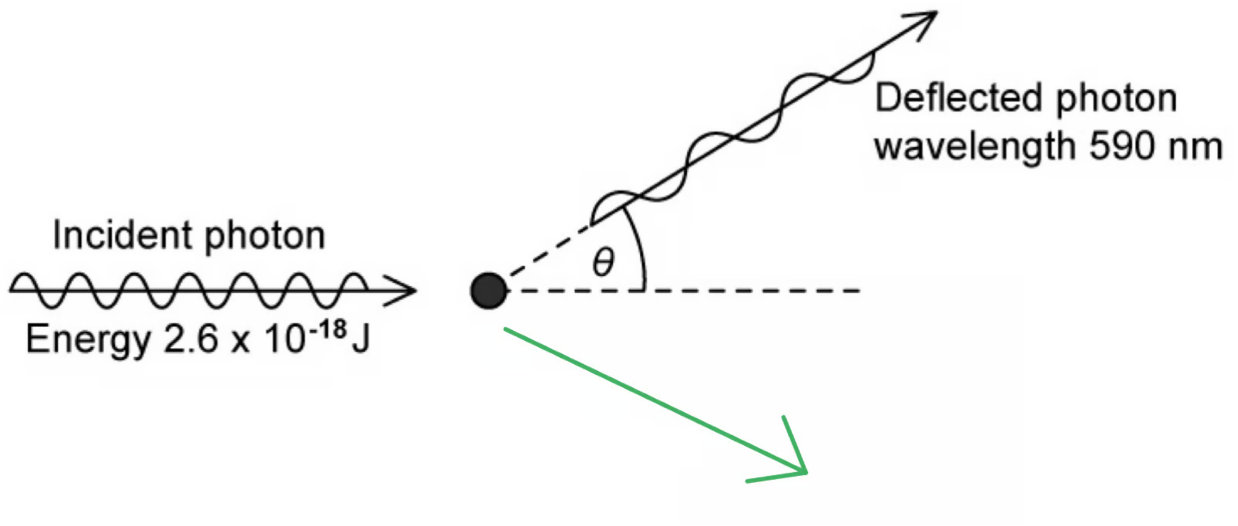
[Total: 1 mark]

3b

(b)

(i) A diagram that indicates a possible initial direction of motion of the electron after the photon has been deflected that would score 1 mark should look like this:

(i) A diagram that indicates a possible initial direction of motion of the electron after the photon has been deflected that would score 1 mark should look like this:



- The arrow drawn should be below the axis and pointing to the right; [1 mark]

(ii) Calculate the energy of the deflected photon:

List the known quantities:

- Planck constant, $h = 6.63 \times 10^{-34}$ J s
- Speed of light in free space, $c = 3.00 \times 10^{8}$ m s^{-1}
- Wavelength of deflected photon, $\lambda = 590$ nm $= 590 \times 10^{-9}$ m

Use the equation for calculating photon energy, E:

- $E = \frac{hc}{\lambda}$
- $E = \frac{(6.63 \times 10^{-34}) \times (3.00 \times 10^{8})}{(590 \times 10^{-9})}$ [1 mark]
- $E = 3.37 \times 10^{-19}$ J [1 mark]

[Total: 3 marks]

Consider the conservation of momentum to figure out the direction of motion of the electron. When resolved:

horizontal component of photon momentum = horizontal component of electron momentum

3c

(c) Calculate the speed of the electron after the photon has been deflected:

List the known quantities:

- Energy of incident photon = 2.6×10^{-18} J (From part (a))
- Energy of deflected photon = 3.37×10^{-19} (From part (b))
- Rest mass of an electron, $m_e = 9.11 \times 10^{-31}$ kg

Calculate the energy of the electron, *E*:

- Energy is conserved, so *E* = energy of incident photon – energy of deflected photon
- $E = (2.6 \times 10^{-18}) - (3.37 \times 10^{-19})$
- $E = 2.26 \times 10^{-18}$ J [1 mark]

Use the de Broglie equation to calculate the speed of the electron, *v*:

- Electron has kinetic energy only after deflection
- So, $E = KE = \frac{1}{2} m_e v^2$
- $v = \sqrt{\frac{2KE}{m_e}}$ [1 mark]
- $v = \sqrt{\frac{2 \times (2.26 \times 10^{-18})}{(9.11 \times 10^{-31})}}$
- $v = 2.23 \times 10^6 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

3d

(d) The magnitude of the final momentum of the electron is not equal to the change in magnitude of the momentum of the photon because:

- Momentum is a vector quantity; [1 mark]

Any **one** from:

- So the momentum must be considered in two directions; [1 mark]
- The direction of the electron changes, so magnitude alone cannot be considered; [1 mark]

[Total: 2 marks]

Read this question carefully! It is asking about the **magnitude** of the momentum.